
AUDITING THE FACTORIZED-SAMPLING CONTRACT: A FOUR-TERM DECOMPOSITION ACROSS MODELS AND KNOWN FACTORS

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ABSTRACT

Independent factorized generation replaces the encoded joint $q(z_c, z_s | y)$ with the product $p(z_c | y)p(z_s)$. We formalize agreement between these distributions as a sampling contract and decompose its latent mismatch exactly into four non-negative terms: global style-prior mismatch, conditional style leakage, semantic-prior mismatch, and within-content coupling. Aggregate prior matching audits only the first clause. We therefore introduce term-aligned, held-out diagnostics and test whether they distinguish practically different trained-model regimes. A smooth construction with $q(z) = \mathcal{N}(0, I)$ exactly supplies the minimal counterexample: global MMD remains calibrated at 6% rejection over 50 replications while a label probe exceeds 90%. In a common-population three-family audit, F-CS-WAE exhibits conditional failure, DRIT fails already at aggregate matching, and DIVA is a near-null boundary case with chance-level residual probes and high semantic retention. Decoder interventions show that the latent distinctions are operational: decoded identity follows the style donor in 99.28% of MNIST and 63.92% of CIFAR-10 swaps, then falls to 23.37% and 35.92% under a conditional intervention. Finally, a three-seed Shapes3D audit finds shape 99.0% recoverable from F-CS-WAE style, whereas a learned content-style VAE remains near the 25% chance level. Thus factorized generation requires a joint distributional contract, not a single favorable marginal statistic, and the proposed audit can diagnose distinct failures as well as informative negative results.

1 INTRODUCTION

Factorized models are sampled from a different joint distribution than the one presented to their decoder during encoding. The encoder induces $q(z_c, z_s | y)$, whereas independent generation draws from $p(z_c | y)p(z_s)$. Treating that replacement as valid therefore makes a specific distributional claim: the encoded joint must agree with the generation-time product, at least along directions used by the decoder.

Aggregate prior matching verifies only one clause of this claim. It can make $q(z_s)$ resemble a Gaussian without establishing content-invariant style, correct class-conditional semantic priors, or conditional independence between the two learned codes. During training, the decoder may exploit semantic and style codes from the same example; at generation, those codes are recombined independently. Reconstruction, semantic clustering, and a favorable global prior statistic can consequently coexist with a failed sampling contract.

This phenomenon is related to content leakage studied in earlier content-style models (Ridgeway & Mozer, 2018); our claim is not that label information in style was previously unknown, nor that the elementary distinction between marginal and conditional distributions is itself the main novelty. We instead characterize the complete joint mismatch implicitly assumed by independent factorized generation, identify which clause aggregate objectives constrain, and turn the remaining clauses into an operational audit. This shift matters: leakage is one possible failure of the sampling contract, but aggregate-prior failure, semantic-prior failure, and within-class coupling are distinct and require different evidence.

We treat this as an audit problem rather than a claim that every split-latent model fails. A population construction first isolates the logical insufficiency of one clause. Trained models then test the more important empirical question: does the term-aligned audit distinguish conditional failure, whole-sale aggregate failure, and an informative near-null result? Decoder interventions test whether the detected structure is used, and Shapes3D supplies explicit generative-factor semantics. Throughout, we keep posterior means, stochastic posterior samples, and prior samples separate: the contract concerns random variables, whereas deterministic swaps commonly use means.

Our contributions are:

1. **Sampling-contract formalization.** We define the mismatch between $q(z_c, z_s | y)$ and $p(z_c | y)p(z_s)$ and decompose it exactly into global style-prior mismatch, conditional style leakage, semantic-prior mismatch, and within-content coupling.
2. **Term-aligned audit.** We specify which held-out diagnostic addresses each clause, distinguish posterior means, posterior samples, and prior samples, and connect latent tests to decoder interventions.
3. **Discriminative evidence.** Across an exact construction, three trained model families, entropy interventions, and known Shapes3D factors, the audit separates conditional failure, aggregate failure, and near-null controls rather than reporting leakage indiscriminately.

2 RELATED WORK

Aggregate matching and disentanglement. VAEs (Kingma & Welling, 2013) penalize a per-example KL, whereas WAEs (Tolstikhin et al., 2017) directly match aggregated posteriors using objectives such as MMD (Gretton et al., 2012). The averaged VAE KL decomposes into an information-capacity term and an aggregate-prior mismatch (Hoffman & Johnson, 2016); neither component alone certifies that a designated style code is independent of a particular label. β -VAE, β -TCVAE, FactorVAE, and DIP-VAE penalize capacity, total correlation, or aggregate moments (Higgins et al., 2017; Chen et al., 2018; Kim & Mnih, 2018; Kumar et al., 2018). These constraints differ from $I(z_s; y)$, and unsupervised disentanglement is not identifiable without assumptions on the model or data (Locatello et al., 2019).

Content-style separation and invariance. Split-latent models use grouped observations, labels, paired supervision, or latent optimization to separate specified and residual variation (Mathieu et al., 2016; Bouchacourt et al., 2018; Jha et al., 2018; Ilse et al., 2020; Gabbay & Hoshen, 2020). Most directly, Ridgeway & Mozer (2018) probe class information in style posteriors and introduce leakage filtering for recombination. Variational fair autoencoders, gradient reversal, Fader Networks, and information-based invariance objectives explicitly suppress designated nuisance information (Louizos et al., 2015; Ganin & Lempitsky, 2015; Lample et al., 2017; Moyer et al., 2018). Our decomposition clarifies why eliminating $I(z_s; y)$ is necessary but does not, by itself, make an independent sampler match the encoded joint.

Recombination. MUNIT and DRIT explicitly combine content with sampled style (Huang et al., 2018; Lee et al., 2018); more recent methods use variance patterns or invertible merging to obtain content-style structure (Wu et al., 2025; Ma et al., 2025). These works focus on learning useful separations. We focus on auditing the probabilistic claim that permits the two learned variables to be sampled independently.

Known-factor evaluation. Class labels are only one semantic partition. Controlled datasets such as Shapes3D expose shape, color, scale, and orientation factors and have been used to evaluate whether representation coordinates track known generative variables (Higgins et al., 2017; Locatello et al., 2019; Google DeepMind, 2018). We use them differently: a designated content/style split is tested in both directions, with family-wise corrected dependence tests and pixel-space swap evaluation.

3 FACTORIZED SAMPLING AND ITS MISMATCH

Let $q(y)$ denote the empirical label distribution and let an encoder induce $q(z_c, z_s | y)$. A factorized class-conditional sampler draws

$$z_c \sim p(z_c | y), \quad z_s \sim p(z_s), \quad (1)$$

independently. In our case $p(z_s) = \mathcal{N}(0, I)$. Many aggregate objectives constrain only

$$\mathcal{L}_{\text{style}} = D(q(z_s), p(z_s)), \quad q(z_s) = \mathbb{E}_{q(y)} q(z_s | y). \quad (2)$$

3.1 THE GENERATION-TIME CONTRACT

The relevant object is not either marginal in isolation, but the discrepancy between the encoded joint and the distribution supplied to the decoder at generation time. We quantify that replacement before asking what any one regularizer can certify.

Theorem 1 (Factorized-sampling contract decomposition). *For any $q(y)$, define*

$$M_{\text{fact}} := \mathbb{E}_{q(y)} [\text{KL}(q(z_c, z_s | y) || p(z_c | y)p(z_s))]. \quad (3)$$

Then

$$\begin{aligned} M_{\text{fact}} = & \underbrace{\text{KL}(q(z_s) || p(z_s))}_{(1) \text{ global style-prior mismatch}} + \underbrace{I_q(z_s; y)}_{(2) \text{ conditional style leakage}} \\ & + \underbrace{\mathbb{E}_{q(y)} \text{KL}(q(z_c | y) || p(z_c | y))}_{(3) \text{ semantic-prior mismatch}} + \underbrace{I_q(z_c; z_s | y)}_{(4) \text{ within-class dependence}}. \end{aligned} \quad (4)$$

All four terms are non-negative.

Corollary 1 (Exact latent sampling-contract match). $M_{\text{fact}} = 0$ iff $q(z_c, z_s | y) = p(z_c | y)p(z_s)$ for $q(y)$ -almost every y . Moreover, for a fixed decoder g ,

$$\mathbb{E}_{q(y)} \text{KL}(g_{\#} q(\cdot | y) || g_{\#} [p(z_c | y)p(z_s)]) \leq M_{\text{fact}}. \quad (5)$$

The inequality is one-way. A non-injective decoder can ignore mismatched latent directions, so latent equality is not necessary for identical decoded distributions; nor does equality between posterior-decoded and prior-decoded outputs establish agreement with real data. Decoder-level evaluation is therefore complementary to the decomposition.

Equation 2 constrains only term 1. Terms 2–4 concern how aggregate mass is partitioned by content, whether the semantic code follows its requested class prior, and whether the two codes remain coupled within a class. They are non-negative but not interchangeable: removing term 2 still permits the decoder to exploit term-4 dependence. Table 1 maps these distinct clauses to the evidence used here.

Table 1: The four latent requirements and their term-aligned evidence. The statistics are proxies for distinct properties, not additive estimates of M_{fact} ; term 3 is not claimed as a calibrated held-out KL test.

Term	Failure source	Required equality	Evidence used here
1	Style-prior mismatch	$q(z_s) = p(z_s)$	global MMD
2	Conditional leakage	$q(z_s y) = q(z_s)$	probe, HSIC, Δ_{inter}
3	Semantic-prior mismatch	$q(z_c y) = p(z_c y)$	training objective; competence
4	Within-class coupling	$q(z_c, z_s y) = q(z_c y)q(z_s y)$	conditional HSIC

3.2 WHY ONE CLAUSE CANNOT CERTIFY THE CONTRACT

The following minimal counterexample is not the primary object of the paper; it isolates why exact satisfaction of term 1 cannot certify term 2, and hence cannot certify the joint contract.

Proposition 1 (No clause-1 certificate for clause 2). *Let $z_s \sim \mathcal{N}(0, I)$ and let $\{A_k\}_{k=1}^K$ be a measurable partition with $P(z_s \in A_k) = 1/K$. Define $y = k$ iff $z_s \in A_k$. Then $q(z_s) = \mathcal{N}(0, I)$ exactly for every marginal divergence D , while $I(z_s; y) = H(y) = \log K$.*

The construction is intentionally extremal, but the issue is not restricted to discontinuous partitions. Let $z \sim \mathcal{N}(0, 1)$ and $P(y = 1 | z) = \sigma(az)$. Symmetry leaves the marginal of z exactly Gaussian, while $q(z | y = 1) = 2\phi(z)\sigma(az)$ and $q(z | y = 0) = 2\phi(z)\sigma(-az)$ are smooth, full-support, and increasingly distinguishable as a grows. Section 5.1 instantiates this construction only to establish clause-wise insufficiency; the trained-model experiments instead evaluate whether the complete audit is operationally discriminative. Proofs are in Appendix A.

4 TERM-ALIGNED AUDIT AND CASE STUDY

4.1 VIEW-SPECIFIC DIAGNOSTICS

We use stochastic posterior samples for claims about the sampling contract and means for deterministic representations:

- **Term 1:** unbiased multiscale-RBF MMD²($\{z_s^i\}, \mathcal{N}(0, I)$).
- **Term 2:** class-mean separation $\Delta_{\text{inter}} = \binom{K}{2}^{-1} \sum_{j < k} \|\bar{z}_s^{(j)} - \bar{z}_s^{(k)}\|_2$, a held-out probe suite, and a multiscale HSIC permutation test between style and label. We compute each on both μ_s and a seeded z_s draw.
- **Term 3:** class-conditional semantic MMD is part of training, and held-out content competence checks guard against semantic collapse. We do not treat either as a calibrated estimate of the KL term.
- **Term 4:** classwise conditional HSIC with within-class permutations, computed on (μ_c, μ_s) and on ten seeded posterior draws (z_c, z_s) per checkpoint. It is a calibrated dependence test, not an estimate of conditional mutual information.

Terms 1, 2, and 4 receive calibrated held-out tests. Term 3 is controlled by the class-conditional semantic objective and checked through competence, but is not interpreted as a calibrated held-out KL estimate. Accordingly, we claim a four-term decomposition with term-aligned diagnostics, not a numerical four-term estimator of M_{fact} .

The fixed audit uses 2,048 test examples and a stratified 60/20/20 probe-train/validation/test split. Feature standardization is fit on probe training data only. We report logistic regression, a fixed two-layer MLP, RBF-SVM, and k -NN; hyperparameters and per-seed values are in Appendix C. Label HSIC uses 1,024 examples; conditional HSIC uses all 2,048. Both use 200 plus-one calibrated permutations, so the smallest attainable p -value is $1/201$. Global MMD uses an independent Gaussian RNG stream and a separate 500-permutation equality calibration.

4.2 CASE-STUDY MODEL

F-CS-WAE uses a shared modified ResNet-18 encoder, a hyperspherical semantic code $z_c \in \mathbb{S}^{63}$, a Gaussian style code $z_s \in \mathbb{R}^{128}$, and a residual upsampling decoder. The encoder trunk feeds separate heads for the semantic direction and concentration (μ_c, ρ_c) and style parameters $(\mu_s, \log \sigma_s^2)$. A forward pass samples both codes and decodes their concatenation. Class-specific Spherical Cauchy priors with EMA-updated centers shape z_c ; global MMD matches sampled z_s to $\mathcal{N}(0, I)$.

The complete objective is

$$\begin{aligned} \mathcal{L} = & \mathcal{L}_{\text{rec}} + \alpha(t)\mathcal{L}_{\text{class}} + \beta(t)\mathcal{L}_{\text{agg}} + \gamma(t)\mathcal{L}_{\text{style}} \\ & + \delta(t)\mathcal{L}_{\text{style-clc}} + \eta(t)\mathcal{L}_{\text{cls}}. \end{aligned} \quad (6)$$

$\mathcal{L}_{\text{rec}}$ combines L1 and LPIPS; the supervised class MMD matches $q(z_c | y = k)$ to its class prior, aggregate semantic MMD matches $q(z_c)$ to the prior mixture, global style MMD implements Eq. 2, and an auxiliary classifier on μ_c prevents semantic collapse. Training lasts 300 epochs in four phases: reconstruction warm-up, style regularization and classification, partial semantic matching, and the full objective. All coefficients and phase boundaries are listed in Appendix B.

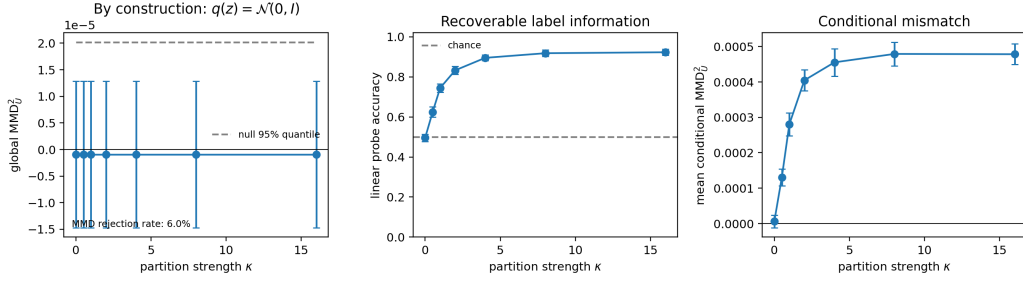


Figure 1: **Exact marginal matching does not certify label independence.** The same Gaussian sample is relabeled as partition strength κ changes. Global MMD therefore remains invariant and rejects in $3/50 = 6\%$ replications, compatible with its nominal 5% level. Meanwhile, probe accuracy exceeds 90% at $\kappa = 8, 16$ and conditional discrepancy increases. Points and bars are mean $\pm$ one standard deviation.

The optional per-class loss

$$\mathcal{L}_{\text{style-cl}} = \frac{1}{K} \sum_k \text{MMD}^2(q(z_s | y = k), \mathcal{N}(0, I)) \quad (7)$$

targets term 2 but not term 4. It reuses labels already consumed by the semantic objective and is therefore a supervised intervention, not a generic unsupervised disentanglement loss. The held-out Stage-0 audit below focuses on unrepaired $\delta = 0$ checkpoints to isolate the failure of marginal matching. We subsequently use existing $\delta > 0$ checkpoints only to test whether explicitly conditional regularization moves the conditional distribution statistics in the predicted direction.

5 EXPERIMENTS

Evidence ladder and reporting rules. We use four complementary settings. The synthetic experiment controls the population marginal exactly. The original image audit covers MNIST, Fashion-MNIST, and CIFAR-10. The native cross-model comparison uses paired $-30^\circ / +30^\circ$ Rotated-MNIST domains and model seeds $\{0, 1, 2\}$. Finally, Shapes3D supplies six known factors: shape is designated as content, and floor hue, wall hue, object hue, scale, and orientation as style (Google DeepMind, 2018). All headline sampling-contract diagnostics use stochastic posterior samples and held-out splits.

Unbiased MMD^2 estimates may be negative under the null and are never clipped. A failure to reject exact equality is not called proof of equality. HSIC uses plus-one permutation calibration; we report domain-level outcomes rather than average p -values. Shapes3D applies Holm correction separately within the six content tests and six style tests. Final epochs and audit populations are fixed without selecting checkpoints on leakage metrics.

5.1 CAN EXACT MARGINAL MATCHING HIDE CONDITIONAL INFORMATION?

The decomposition makes aggregate matching a clause rather than a certificate. We first isolate that logical point by instantiating the smooth construction from Proposition 1 in 128 dimensions:

$$z \sim \mathcal{N}(0, I), \quad p(y = 1 | z) = \sigma(2\kappa z_1). \quad (8)$$

The latent is sampled before the label, so $q_\kappa(z) = \mathcal{N}(0, I)$ exactly for every κ : term 1 is zero by the data-generating process, not because a test fails to reject. We use 50 replications of 2,048 samples, 499 MMD and HSIC permutations, and the same stratified 60/20/20 linear-probe protocol as the checkpoint audit.

Probe accuracy rises from $49.58 \pm 1.86\%$ at $\kappa = 0$ to $92.30 \pm 1.51\%$ at $\kappa = 16$. HSIC rejects in 100% of replications for every $\kappa > 0$, while its null rejection rate is 2%. Mean conditional MMD rises from 6.16×10^{-6} to 4.78×10^{-4} . The preregistered eight-part composite gate did not fully pass: the requirement of probe accuracy above 90% at each of $\kappa \in \{4, 8, 16\}$ missed at $\kappa = 4$ (89.51%, a

0.49-point gap). All calibration and trend checks passed. We preserve this failure and use only the supported existential claim: exact marginal equality can coexist with more than 90% recoverable label information.

A sweep over $n = 512\text{--}4096$ preserves 92.0–92.5% probe accuracy and 5–6% global-MMD rejection (Appendix 13). This stability is the expected pattern, not a failure of the sample-size study. At fixed κ , the population relation between z_1 and y —and therefore the attainable classification accuracy—does not strengthen when more observations are drawn. Increasing n should mainly stabilize its finite-sample estimate. More importantly, the marginal remains exactly Gaussian for every n , so a valid level-.05 global test should continue to reject only at roughly its nominal rate. The sweep exhibits both properties: the predictive effect is essentially unchanged and the marginal test does not acquire spurious power against a null that is true by construction.

5.2 DOES THE AUDIT DISTINGUISH QUALITATIVELY DIFFERENT FAILURES?

The construction establishes possibility, but learned models serve a different role: they test whether the audit reports leakage indiscriminately or separates practically different regimes. We train native-family, common-backbone adaptations of F-CS-WAE, DRIT (Lee et al., 2018), and DIVA (Ilse et al., 2020) for 100 epochs on the same two-domain Rotated-MNIST population. The audited style variables are F-CS-WAE z_s , DRIT’s domain-specific attribute code, and DIVA’s residual z_x . DRIT domains are audited separately because their attribute spaces are domain-specific.

The audit separates three qualitatively different outcomes. F-CS-WAE has a roughly tenfold conditional/global discrepancy and high, seed-stable label recovery. DRIT’s global discrepancy is already large, so its label-bearing attribute code is a wholesale prior-sampler failure rather than a matched-marginal paradox. DIVA has near-null global discrepancy, chance-level probes, and competent prior-style retention. HSIC rejects in all six seed-domain tests for F-CS-WAE and DRIT ($p = 1/201$). Two of six raw DIVA tests fall below .05, but none survives Holm correction within the DIVA family and the probe pattern does not repeat. We therefore say that the audit finds no stable, practically recoverable DIVA residual leakage at this power—not that independence has been proven.

The frozen all-check gate is intentionally shown even though its retention component is also an outcome of the sampling contract. Post hoc, all three families pass the separable reconstruction/content checks, while only DIVA passes the full preregistered gate. DRIT and DIVA are faithful native-objective common-backbone adaptations, not exact reproductions of the authors’ benchmark architectures; this experiment supports audit discrimination, not a ranking of the original methods.

5.3 IS LEAKAGE ONLY A CONSEQUENCE OF POSTERIOR COLLAPSE?

F-CS-WAE’s raw variance head often reaches its numerical floor. To test whether that alone creates leakage, we retrain MNIST seed 0 and CIFAR-10 seeds $\{0, 1, 2\}$ with

$$\sigma_{\text{eff}}^2(x) = \exp(\text{clamp}(\log \sigma_s^2(x), -10, 10)) + \sigma_{\text{floor}}^2, \quad (9)$$

using the same sampler during training and audit. Table 3 reports the complete fixed CIFAR-10 suite and representative MNIST points from its five-level sweep.

On MNIST, increasing entropy by 3.96 nats/dimension still leaves 95.61% label accuracy, while reconstruction loss remains only $1.32\times$ baseline. On CIFAR-10, $\sigma_{\text{floor}} = .15$ is a non-degenerate operating point: the mean probe remains 19.0 points above chance, conditional MMD is $5.3\times$ global MMD, reconstruction increases by about 13.5%, and Gen-ACC improves. At .35, however, the mean linear probe is 10.5%, effectively chance, even though every seed still rejects HSIC and conditional MMD remains positive. We therefore do not call this strong recoverable label leakage. The replicated boundary shows that added stochasticity attenuates the failure without making near-determinism a necessary condition for it. MNIST remains a single-seed intervention; the raw variance head remains floor-seeking.

The high-noise CIFAR-10 result illustrates why test outcomes and effect sizes must be reported together. HSIC asks whether there is evidence of *any* dependence detectable by its kernel, whereas the logistic probe asks whether label information is usefully recoverable through a particular linear

Table 2: Native cross-model audit, mean $\pm$ sample standard deviation over three model seeds. Probe and retention entries are percentages. “Base” checks reconstruction and content representation; “all” additionally includes the preregistered 70% prior-style retention gate.

Family	Global MMD	Cond. MMD	Logit z_s	RBF z_s	Content	Rec. L1	Prior retain	Base / all
F-CS-WAE	.003326 $\pm$.000140	.032262 $\pm$.001197	91.92 $\pm$ 2.43	96.67 $\pm$.95	99.42 $\pm$.14	.01510 $\pm$.00036	12.37 $\pm$ 1.19	3/3/0/3
DRIT	.255261 $\pm$.008545	.272442 $\pm$.001938	34.42 $\pm$ 12.62	32.50 $\pm$ 14.12	88.25 $\pm$ 1.52	.04284 $\pm$.00211	9.87 $\pm$.12	3/3/0/3
DIVA	-.0000179 $\pm$.0000274	.000176 $\pm$.000129	10.67 $\pm$ 1.04	11.83 $\pm$.63	96.67 $\pm$.76	.03449 $\pm$.00009	89.45 $\pm$.83	3/3/3/3

Table 3: Entropy intervention. MNIST is seed 0; CIFAR-10 entries are rounded three-seed means. Probe/Gen-ACC are percentages; every CIFAR seed rejects HSIC at $p = 1/201$.

Data	σ_{floor}	H	Probe	Global MMD	Cond. MMD	HSIC p	Gen-ACC	Rec.
MNIST	0	-3.581	99.76	.000579	.016024	.00498	10.15	.01259
MNIST	.15	-.477	98.78	.000654	.012044	.00498	37.35	.01520
MNIST	.35	.381	95.61	.000518	.007516	.00498	93.50	.01658
CIFAR-10	0	-3.581	68.0	.00030	.0058	.00498	13.0	.052
CIFAR-10	.15	-.477	29.0	.00070	.0037	.00498	27.0	.059
CIFAR-10	.35	.369	10.5	.00067	.0013	.00498	52.0	.065

decision rule. Consequently, $p = .00498$ at $\sigma_{\text{floor}} = .35$ does not contradict a 10.5% probe: the representation can remain statistically non-independent while carrying little linearly exploitable label signal. The simultaneous fall in conditional MMD from .0058 to .0013 supports attenuation, rather than an estimator-only explanation. Gen-ACC rises as this mismatch falls, but because every level is separately trained, we treat that alignment as mechanism evidence rather than a causal estimate of the effect of conditional MMD alone.

5.4 DOES THE LATENT MISMATCH AFFECT THE DECODER?

A recoverable label in style matters operationally only if the decoder uses it. We therefore pair the original fixed audit on MNIST, Fashion-MNIST, and CIFAR-10 with prior-sampling and donor-swap interventions. Exact global equality is rejected for every learned checkpoint, so “small” below is only relative to conditional discrepancy; these models are not used to establish the exact-marginal counterexample.

Across the new three-seed aggregates, conditional MMD is .0160 on MNIST and .010 on Fashion-MNIST; all seedwise HSIC tests attain $p = 1/201$. The post-hoc class-style sampler is fitted on encoded training styles and is a mechanism check, not a factorization repair. Its gain shows that the decoder is sensitive to the conditional mismatch. Deterministic off-diagonal mean swaps agree: decoded identity follows the style donor in 99.28% of MNIST and 63.92% of CIFAR-10 cases, versus the semantic donor in .06% and 4.72%.

Archived seed-0 checkpoints provide a directional conditional intervention. Adding the per-class style objective in Eq. 7 reduces style-donor following to 23.37% on MNIST and 35.92% on CIFAR-10 while increasing semantic-donor following. Yet conditional HSIC still rejects on all ten stochastic posterior draws at $\delta = 1$. This does not prove that fixing term 2 can never fix sampling; rather, the joint pattern is consistent with Theorem 1: suppressing class-dependent style structure can improve decoder behavior while leaving detectable term-4 coupling. Because the intervention is one seed, we treat it as mechanism evidence, not a remedy benchmark. Full probes, conditional MMD, and swap matrices are retained in Appendix D–E.

Table 4: Original F-CS-WAE audit. MMD/probe entries are rounded three-seed aggregates; generation retains its original scope (MNIST seed 0, CIFAR three seeds). Every seed rejects HSIC at $p = 1/201$.

Dataset	Global MMD	Logit z_s	Global Gen-ACC	Class-style Gen-ACC
MNIST	.00058	99.6	.1047	1.0000
Fashion-MNIST	.00034	88.0	—	—
CIFAR-10	.000298 ± .000052	68.62 ± 6.19	.1283 ± .0032	.4716 ± .0686

5.5 DOES THE PATTERN EXTEND TO EXPLICIT GENERATIVE FACTORS?

Shapes3D contains all combinations of six independent factors. We train F-CS-WAE, an observed-shape conditional VAE (Sohn et al., 2015), and a learned supervised content–style VAE with class-conditional content priors (Ilse et al., 2020). All use a native 64×64 decoder and the same 120k/20k/20k shape-stratified IID split at fixed epoch 100. F-CS-WAE and the learned VAE use seeds $\{0, 1, 2\}$; the oracle observed-shape cVAE remains the predefined seed-0 control. An independent multi-head image evaluator passes every preregistered factor threshold, and all three generators pass the reconstruction/content competence gate.

Table 5: Shapes3D audit. Except for seed-0 global MMD, F-CS-WAE/learned-VAE entries are rounded three-seed means; the observed-shape cVAE is seed 0. $z_c \rightarrow \text{style}$ averages five nuisance-factor probes.

Model	Global MMD (p)	Cond. shape MMD	$z_s \rightarrow \text{shape}$	$z_c \rightarrow \text{style}$	Prior shape retain	Swap shape retain	Rec. L1
F-CS-WAE	.000766 (.002)	.0085	99.0	87.0	89.0	88.0	.0023
Observed-shape cVAE	.000026 (.092)	.000076	25.98	10.17	99.88	100.00	.00260
Learned C/S VAE	— .0000005 (.501)	.00004	25.5	9.5	99.7	99.8	.0029

Across three seeds, F-CS-WAE style predicts shape at 99.0% and its content predicts nuisance factors at 87.0%. At seed 0, HSIC rejects all six factor tests in each view after Holm correction. By contrast, both controls place style-to-shape probes near chance and retain no Holm-significant shape dependence in style; their content codes reject only for shape, while their style codes reject for the five intended nuisance factors. Across three seeds, prior-style and posterior-swap decodes preserve shape at 99.7% and 99.8% for the learned control but only 89.0% and 88.0% for F-CS-WAE. Thus the class-label finding is not merely an arbitrary semantic proxy in this controlled setting: the designated shape factor cross-loads into style, while nuisance factors cross-load into content.

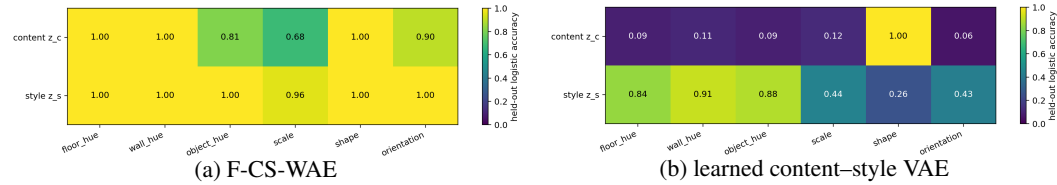


Figure 2: Held-out factor loading on Shapes3D stochastic latents. F-CS-WAE stores nearly every factor in both codes; the learned VAE places shape in content and nuisance factors in style. Accuracy is descriptive; the text reports Holm-corrected HSIC.

The F-CS-WAE/learned-control contrast repeats over three predefined seeds on an IID factor split. It establishes seed-level replication under that split, not compositional generalization to unseen factor combinations; the observed-shape oracle remains a single-seed auxiliary control.

The controls are essential to this interpretation. A probe near 25% on the four-way shape task shows that the audit does not automatically extract shape from every style code, while 99.7–99.8% retention shows that the same evaluator can recognize preserved shape after prior sampling and swaps. F-CS-WAE instead combines a 99.0% style-to-shape probe with only 89.0% prior and 88.0% swap retention. Thus the latent diagnostic and the decoded intervention point in the same direction: shape is cross-loaded into style, and breaking the encoded pairing changes shape more often. The

learned VAE is not proof of perfect disentanglement, but it is a concrete negative control showing that this protocol can recover the intended split.

6 DISCUSSION AND LIMITATIONS

6.1 WHAT THE CONTRACT AND EVIDENCE ESTABLISH

The central result is the contract, not the observation that marginals and conditionals differ. Theorem 1 characterizes exactly what is changed when an encoded pair from $q(z_c, z_s | y)$ is replaced by an independent draw from $p(z_c | y)p(z_s)$. Proposition 1 and the synthetic experiment then serve the narrower purpose of showing that perfect clause-1 satisfaction leaves clause 2 unconstrained.

The learned-model evidence evaluates whether this formal distinction is operationally useful. It supports three conclusions. First, F-CS-WAE shows conditional failure across original image checkpoints, three controlled Rotated-MNIST seeds, a three-seed CIFAR-10 entropy intervention, and a three-seed known-factor Shapes3D replication. Second, the audit is discriminative rather than universally positive: it separates that regime from DRIT’s aggregate-prior failure and DIVA’s near-null boundary case, while the Shapes3D VAE controls place the designated semantic factor in the intended code. Third, prior sampling and donor swaps connect latent findings to decoder behavior instead of assuming every detectable mismatch must affect pixels.

None of these results says that every factorized model leaks, that a non-rejection proves independence, or that probe accuracy equals mutual information. Likewise, a decoder can ignore a latent mismatch, so the four-term KL decomposition is an upper-level contract, not a claim that every nonzero latent statistic must visibly corrupt pixels. This is why the audit pairs prior and conditional statistics with independent pixel evaluators and interventions.

A minimal sampling-contract audit. For a proposed sampler, report: (i) the encoded joint and generation-time product; (ii) global prior fit on stochastic posterior samples; (iii) held-out conditional probes and a calibrated dependence test; (iv) within-content dependence; and (v) prior or swap outcomes scored by an independent evaluator. Name posterior means, posterior samples, and prior samples explicitly. On known-factor data, audit both directions: content should not predict designated style factors, nor style predict designated content. The cross-model and entropy results show why the full pattern is preferable to one scalar score: aggregate failure, recoverable leakage, weak residual dependence, and a near-null result call for different conclusions.

6.2 LIMITATIONS AND THREATS TO VALIDITY

The scope remains bounded. The MNIST entropy sweep and the observed-shape Shapes3D oracle use one training seed; the CIFAR-10 entropy sweep and the F-CS-WAE/learned-VAE Shapes3D comparison use three. Shapes3D uses an IID shape-stratified split rather than unseen combinations. DRIT and DIVA are native-objective common-backbone adaptations on one controlled two-domain dataset, not official benchmark reproductions. The Shapes3D pixel evaluator may share inductive biases with the audited encoders despite being trained independently. Decoder interventions on MNIST remain seed 0. The externally completed follow-ups are available here as rounded aggregate exports, so we do not infer missing standard deviations from headline means.

MMD and HSIC remain kernel- and power-dependent finite-sample statistics, not KL or mutual-information estimates. Holm correction controls the declared Shapes3D and DIVA test families but does not turn failure to reject into equivalence. Finally, labels and Shapes3D factors are controlled definitions of content; conclusions need not transfer to dependent factors in natural data.

7 CONCLUSION

Independent factorized generation makes a joint claim: the encoded $q(z_c, z_s | y)$ must be replaceable by $p(z_c | y)p(z_s)$. The four-term decomposition exposes its clauses; term-aligned diagnostics distinguish aggregate failure, conditional leakage, coupling, and near-null results. The exact construction establishes clause-wise insufficiency, while trained-model and decoder evidence establish its operational value.

REPRODUCIBILITY STATEMENT

Section 4 specifies the core latent views, preprocessing, probe selection, and permutation calibration. Sections 5.1–5.5 state the setting-specific populations, seeds, and fixed reporting rules. Appendix A contains complete proofs; Appendices C–E retain the original checkpoint audit; and Appendices G–J provide the expanded protocols, per-level/per-seed results, multiplicity rules, competence gates, and artifact boundaries. Unrounded seed-0 and cross-model values are tied to JSON manifests; newly completed sample-size and three-seed follow-ups are identified as rounded aggregates. An anonymous supplementary archive will include the manifests, follow-up aggregate exports, frozen criteria, and figure sources.

ETHICS STATEMENT

This work uses standard public computer-vision benchmarks and existing model checkpoints, collects no new data, and involves no human subjects or sensitive personal information. Its purpose is to identify and measure a failure mode in factorized generative models. Generated examples are low-resolution benchmark images; nevertheless, the broader recommendation—auditing whether a decoder uses unintended latent information—may also help identify representation leakage in higher-stakes applications.

AI USE STATEMENT

Generative AI tools were used to provide feedback on evaluation methodology and experimental design, assist with interpretation of diagnostic results, audit consistency between implementation artifacts and reported procedures, and help restructure and edit the manuscript. They were not used to generate datasets or experimental measurements. The authors inspected the cited literature, verified the mathematical arguments, executed and tested the code, checked the experimental artifacts and every reported numerical value, and reviewed all AI-assisted text. The authors take responsibility for the final content of this work.

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A PROOFS

A.1 PROOF OF PROPOSITION 1

The label is defined after drawing z_s and therefore does not alter the marginal law of z_s . Hence $q(z_s) = \mathcal{N}(0, I)$ and any divergence between these two identical measures is zero. Because y is a deterministic function of z_s , $H(y | z_s) = 0$. The equal-probability partition gives $H(y) = \log K$, and consequently $I(z_s; y) = H(y) - H(y | z_s) = \log K$.

A.2 A SMOOTH, FULL-SUPPORT CONSTRUCTION

Let $z \sim \mathcal{N}(0, 1)$ and, conditional on z , draw a binary label with

$$P(y = 1 | z) = \sigma(az), \quad P(y = 0 | z) = \sigma(-az), \quad (10)$$

where $a > 0$ and $\sigma(t) = (1 + e^{-t})^{-1}$. Since the standard normal density ϕ is symmetric and $\sigma(t) + \sigma(-t) = 1$,

$$P(y = 1) = \int_{\mathbb{R}} \phi(z) \sigma(az) dz \quad (11)$$

$$= \frac{1}{2} \int_{\mathbb{R}} \phi(z) [\sigma(az) + \sigma(-az)] dz = \frac{1}{2}. \quad (12)$$

The generative construction starts with the standard normal and changes only the conditional label law, so the marginal of z remains exactly standard normal. Bayes' rule gives

$$q(z | y = 1) = 2\phi(z)\sigma(az), \quad q(z | y = 0) = 2\phi(z)\sigma(-az). \quad (13)$$

Both densities are smooth and strictly positive on $\mathbb{R}$. When $a \rightarrow \infty$, the conditional label approaches the indicator of the sign of z outside a set of Gaussian measure zero. Thus $H(y | z) \rightarrow 0$ and $I(z; y) \rightarrow \log 2$ while the marginal remains unchanged.

A.3 PROOF OF THEOREM 1

For a fixed y , insert the product of the encoder conditionals and apply the KL chain rule:

$$\text{KL}(q(z_c, z_s | y) \| p(z_c | y)p(z_s)) \quad (14)$$

$$= \text{KL}(q(z_c, z_s | y) \| q(z_c | y)q(z_s | y)) \\ + \text{KL}(q(z_c | y) \| p(z_c | y)) + \text{KL}(q(z_s | y) \| p(z_s)). \quad (15)$$

The first term is $I_q(z_c; z_s | y)$ at that label. Averaging over $q(y)$ gives term 4 and term 3 of Eq. 4. For the last term, insert the aggregate posterior $q(z_s)$:

$$\mathbb{E}_{q(y)} \text{KL}(q(z_s | y) \| p(z_s)) = \mathbb{E}_{q(y)} \text{KL}(q(z_s | y) \| q(z_s)) \\ + \text{KL}(q(z_s) \| p(z_s)) \quad (16)$$

$$= I_q(z_s; y) + \text{KL}(q(z_s) \| p(z_s)). \quad (17)$$

This establishes the equality for arbitrary $q(y)$. Non-negativity follows from non-negativity of KL divergence and mutual information.

A.4 PROOF OF COROLLARY 1

An expectation of a non-negative KL is zero iff its integrand is zero for $q(y)$ -almost every y , which holds iff the two latent conditional measures are equal. The pushforward inequality follows by applying the data-processing inequality to the deterministic Markov kernel induced by the decoder for each y , then averaging over $q(y)$.

B ARCHITECTURE AND TRAINING

B.1 ARCHITECTURE

The shared encoder is a ResNet-18 adapted to small images: its first 7×7 , stride-2 convolution is replaced by a 3×3 , stride-1 convolution and max pooling is removed. The trunk output is flattened and mapped to 256 dimensions with SiLU. Separate heads produce an unnormalized semantic direction, a scalar semantic concentration, a style mean, and a style log variance. We use

$$\mu_c = \widetilde{\mu}_c / \|\widetilde{\mu}_c\|_2, \quad \rho_c = \sigma(r_c)(1 - \varepsilon), \quad (18)$$

$$z_s = \mu_s + \exp(\frac{1}{2} \log \sigma_s^2) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I). \quad (19)$$

The semantic code is sampled by a Spherical Cauchy Möbius reparameterization. The decoder concatenates $[z_c; z_s]$, projects it to a $512 \times 4 \times 4$ tensor, and applies three residual bilinear-upsample blocks with channel widths 256, 128, 64. Its sigmoid output is interpolated to the native dataset resolution when necessary. MNIST and Fashion-MNIST therefore remain 28×28 and single-channel; CIFAR-10 remains $32 \times 32 \times 3$.

For class k , the semantic prior is a Spherical Cauchy distribution centered at an EMA direction m_k , with concentration $\rho_p = 0.7$. The center is updated once per epoch from encoded class-mean directions with momentum 0.95. The style prior is $\mathcal{N}(0, I)$.

B.2 OBJECTIVE

The training objective is

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \alpha(t)\mathcal{L}_{\text{class}} + \beta(t)\mathcal{L}_{\text{agg}} + \gamma(t)\mathcal{L}_{\text{style}} \\ + \delta(t)\mathcal{L}_{\text{style-cl}} + \eta(t)\mathcal{L}_{\text{cls}}. \quad (20)$$

$\mathcal{L}_{\text{rec}}$ is L1 plus 0.1 LPIPS (Zhang et al., 2018). $\mathcal{L}_{\text{class}}$ matches each semantic class conditional to its spherical prior, $\mathcal{L}_{\text{agg}}$ matches aggregate semantic codes to the class-prior mixture, $\mathcal{L}_{\text{style}}$ matches sampled style codes to a standard Gaussian, and $\mathcal{L}_{\text{cls}}$ is cross entropy from μ_c . The baseline checkpoints audited in the main paper explicitly set $\delta = 0$; the repository default for new remedied runs is $\delta = 1$, so the run manifest or command-line override is required to distinguish them.

Training has four phases: reconstruction only for epochs 0–49; global style MMD and the auxiliary classifier ramp during epochs 50–99; semantic class and aggregate MMD ramp to half strength during epochs 100–199; all terms reach full strength during epochs 200–299. For remedied runs, per-class style MMD follows the global-style ramp. No image augmentation is used.

Table 6: Case-study training hyperparameters.

Parameter	Value
Optimizer	Adam, learning rate 10^{-3}
Schedule	StepLR, step 100, factor 0.5
Batch size / epochs	128 / 300
Gradient clipping	global norm 1.0
Semantic / style dimension	64 / 128
Semantic-prior concentration	0.7
EMA momentum	0.95
Final class / aggregate MMD weights	2.0 / 5.0
Final global style MMD weight	1.0
Final auxiliary-classifier weight	0.3

C CANONICAL AUDIT PROTOCOL

C.1 DATA SPLIT AND LATENT VIEWS

For each checkpoint we select 2,048 examples from the standard test split with evaluation seed 0. Probe indices are generated by a deterministic stratified 60/20/20 split, yielding 1,228 training, 410 validation, and 410 test examples. Standardization parameters are estimated only on the 1,228 probe-training examples. The split indices are stored by SHA-256 hash in each result manifest.

The encoder is evaluated once to obtain $(\mu_c, \mu_s, \log \sigma_s^2)$ and a seeded stochastic draw z_s . We use:

- μ_s for representation probes and deterministic mean swaps;
- z_s for global MMD, conditional style statistics, and the sampling-contract probe; and
- both (μ_c, μ_s) and repeated posterior draws (z_c, z_s) for the conditional-dependence diagnostic.

Agreement between these views is reported as an empirical result rather than assumed.

C.2 PROBE SUITE

The suite contains multinomial logistic regression, a fixed two-layer MLP, RBF-SVM, and k -NN. Validation data are used for fixed early-stopping or model-selection rules; every headline accuracy is evaluated once on the untouched test partition. We report accuracy and macro-F1. The main paper uses logistic regression as the simplest standardized headline and includes RBF-SVM to detect nonlinear separation.

C.3 MMD AND HSIC

Global MMD is the unbiased U-statistic with the averaged Euclidean RBF kernel

$$k(u, v) = \frac{1}{7} \sum_{\sigma \in \{0.5, 1, 2, 5, 10, 20, 50\}} \exp\left(-\frac{\|u - v\|_2^2}{2\sigma^2}\right). \quad (21)$$

The Gaussian reference has the same sample count as the encoded set and uses an RNG stream independent of posterior sampling. A balanced pooled-label permutation test calibrates the finite-sample null with 500 permutations. The resulting null 95th percentiles are $1.88\text{--}2.45 \times 10^{-5}$, whereas observed global MMD is $2.48\text{--}5.79 \times 10^{-4}$; exact equality is rejected for every checkpoint ($p = 1/501$). We therefore use “small” only as a relative numerical description, not as acceptance of the equality null. Δ_{inter} is not an MMD; it is the average Euclidean distance between all pairs of class-conditional style means.

HSIC centers a multiscale style Gram matrix and a label Gram matrix and uses 1,024 examples for bounded memory. We calibrate it by 200 label permutations. With the plus-one correction,

$$p = \frac{1 + \#\{T_b \geq T_{\text{obs}}\}}{201}, \quad (22)$$

so $p = 0.004975$ is reported rather than $p = 0$. Conditional HSIC for term 4 permutes styles independently inside every represented class.

We do not report the current JointMMD U-statistic as a headline result. Its joint and recombined samples share observations, so the nominal independent two-sample U-statistic assumptions require a sample-splitting calibration that is not yet part of the canonical protocol.

D FULL STAGE-0 RESULTS

Table 7: Full held-out probe suite from the original audit. Values are test accuracy in percent. MNIST/Fashion-MNIST rows are seed-0 details; CIFAR-10 is mean $\pm$ sample standard deviation over model seeds 0, 1, 2.

Dataset	View	Logistic	MLP	RBF-SVM	k -NN
MNIST	μ_s	99.76	99.51	99.76	97.80
	z_s	99.76	99.51	99.76	97.80
Fashion	μ_s	88.29	90.73	93.17	85.12
	z_s	88.29	90.73	93.17	85.61
CIFAR-10	μ_s	68.46 ± 6.29	70.49 ± 3.83	70.24 ± 3.23	42.52 ± 6.93
	z_s	68.62 ± 6.19	70.57 ± 3.50	70.16 ± 3.36	42.20 ± 7.11

Table 8: Completed three-seed headline replication for the original MNIST and Fashion-MNIST checkpoints. Values are rounded aggregate results; every seedwise HSIC test attains $p = 1/201$.

Dataset	Global MMD	Logit z_s (%)	Cond. MMD	HSIC p
MNIST	.00058	99.6	.0160	.00498
Fashion-MNIST	.00034	88.0	.010	.00498

Table 9: Per-seed CIFAR-10 results for the two principal probes.

Seed	$\Delta_{\text{inter}}(z_s)$	Logit μ_s	Logit z_s	RBF μ_s	RBF z_s
0	3.801	61.46	61.71	66.59	66.34
1	4.264	73.66	73.66	72.68	72.68
2	3.797	70.24	70.49	71.46	71.46

For term 4, ten stochastic posterior draws per checkpoint all attain the minimum calibrated $p = 1/201$. The sample-view statistic is 0.002682 ± 0.000005 across draws on MNIST, 0.002459 on Fashion-MNIST, and has per-model means 0.002250 , 0.002292 , and 0.002338 on CIFAR-10. The seed-0 $\delta = 1$ checkpoints also reject on all ten draws, with means 0.002507 (MNIST) and 0.002283 (CIFAR-10). These calibrated kernel tests are evidence of stochastic within-class dependence, not numerical estimates of $I(z_c; z_s | y)$.

D.1 STRUCTURE OF THE CONDITIONAL MISMATCH

The class averages reported for the original checkpoints compress a structured ten-class discrepancy. Figure 3 compares every pair of class-conditional style distributions. Broad off-diagonal mass is visible on both datasets, rather than separation being confined to a few class pairs. These matrices use the same stochastic-view diagnostic as the per-class prior comparison; their scale should not be compared across datasets as if it were a normalized dependence measure.

Figure 4 gives a complementary one-dimensional view. The projection direction is computed from the class means rather than chosen after inspecting individual examples. Several MNIST classes separate along this direction, but the overlap also illustrates why a single projection is not a substitute for the held-out multivariate probes.

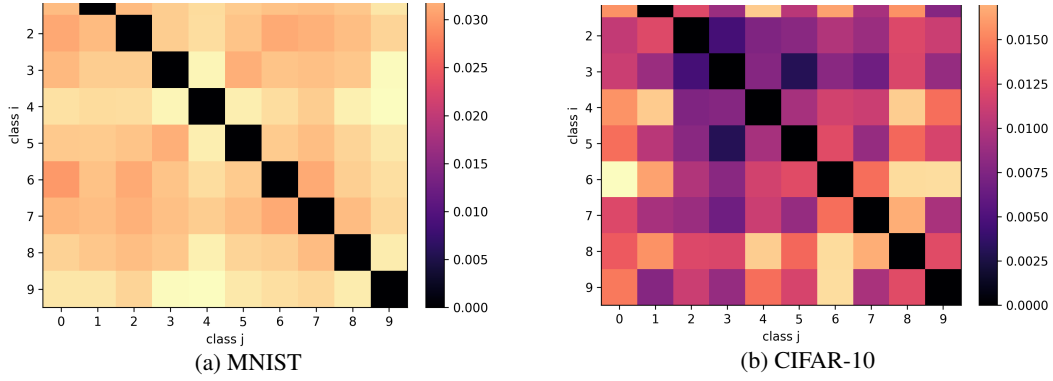


Figure 3: Pairwise $\text{MMD}^2(q(z_s | y = i), q(z_s | y = j))$ for the unrepaired seed-0 checkpoints. The diagonal is zero by construction; off-diagonal structure shows that conditional style distributions differ throughout the label set.

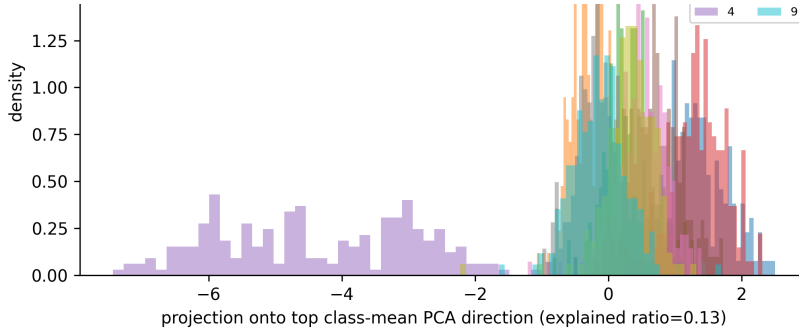


Figure 4: MNIST stochastic styles projected onto the leading direction of the class-mean matrix at $\delta = 0$. The direction explains 13.1% of the class-mean variance. The plot is descriptive; the held-out probes and HSIC tests provide the quantitative evidence.

D.2 POSTERIOR-VARIANCE INSPECTION

The fractions of raw style log-variance coordinates at or below the clamp floor are 99.9966% (MNIST), 99.9989% (Fashion-MNIST), and 100% for all three CIFAR-10 checkpoints. Because this statistic was added during the post-hoc checkpoint audit, it is reported in the human-readable audit report; future manifests should also store it as a first-class numeric field.

D.3 COMPLETE CONDITIONAL-REGULARIZATION SWEEP

Table 10 expands the four points shown in the main paper to all six archived δ settings. We retain ACC, Δ_{inter} , and FID because they are directly tied to the sweep artifacts, but omit the historical probe column until it is regenerated with the canonical held-out split. Each point is a separate single-seed training run.

Table 10: Complete single-seed per-class-style-MMD sweep. Lower FID and Δ_{inter} are better; no uncertainty or significance claim is made.

δ	MNIST			CIFAR-10		
	ACC	Δ_{inter}	FID	ACC	Δ_{inter}	FID
0	.9915	6.21	74.0	.8129	3.74	80.5
.03	.9944	3.27	62.8	.8111	2.48	79.1
.1	.9898	2.12	52.6	.8100	2.03	77.5
.3	.9936	1.55	59.4	.7148	1.59	72.0
1	.8592	1.22	43.4	.8000	1.05	74.4
3	.9929	1.18	41.5	.7994	.95	76.1

E GENERATION AND DECODER DIAGNOSTICS

E.1 EXTERNAL GENERATION EVALUATION

The independent MNIST evaluator is a four-convolution CNN trained for 20 epochs on real images only. Its standard test accuracy is 99.34%. The CIFAR-10 evaluator is a WideResNet-28-10 trained for 200 epochs with standard random-crop and horizontal-flip augmentation on real training images only; its test accuracy is 96.16%. Neither evaluator contributes to the generative model’s loss.

Table 11: Per-checkpoint external Gen-ACC. Each entry uses 1,000 images per requested class and a class-conditional style bank fitted on 12,000 training examples.

Dataset	Model seed	Global Gaussian	Class diagonal $\tau = .25$
MNIST	0	0.1047	1.0000
CIFAR-10	0	0.1262	0.4016
CIFAR-10	1	0.1320	0.5386
CIFAR-10	2	0.1268	0.4747

The class-diagonal distribution is fit from encoded training styles and therefore deliberately conditions style on the requested label. Its gain tests whether the decoder is sensitive to the conditional mismatch. It is not an unconditional generative model and does not repair the encoder.

Figure 5 displays the qualitative counterpart of Table 11. Conditioning style makes the requested category more visually consistent in some rows, but shrinking the diagonal Gaussian to $\tau = .25$ also reduces variation and can produce blurry or prototype-like outputs. This is why Gen-ACC is not interpreted as a stand-alone measure of generative quality; diversity-sensitive evaluation on the same generated pools remains necessary.

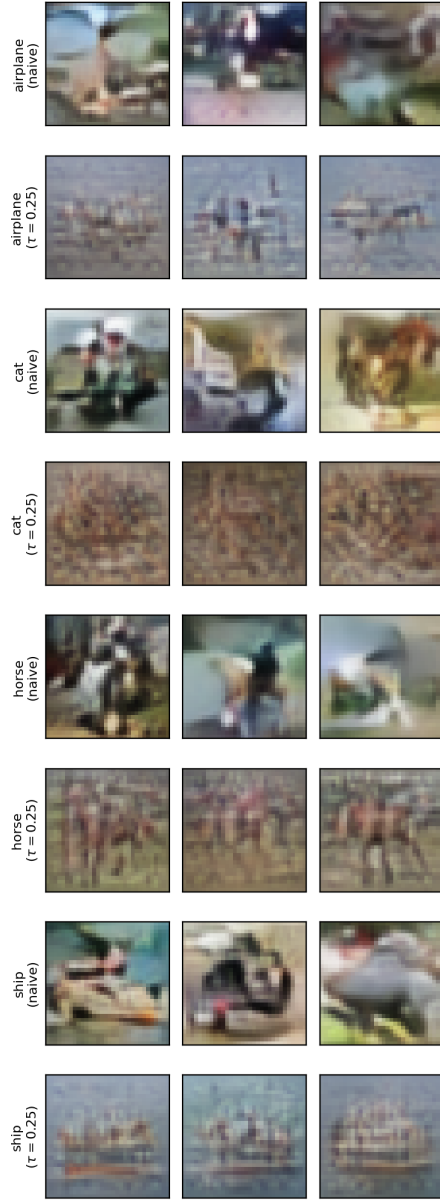


Figure 5: Archived CIFAR-10 samples from the seed-0 checkpoint. For four requested classes, odd rows use the global Gaussian style prior and even rows use the post-hoc class-diagonal prior at $\tau = .25$. The grid is qualitative; external Gen-ACC is reported in Table 11.

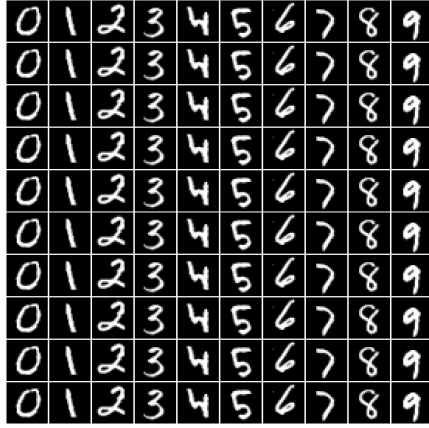
E.2 INDEPENDENT-EVALUATOR DETERMINISTIC MEAN SWAPS

For each ordered pair of distinct classes (a, b) , the diagnostic decodes $\text{Dec}(\mu_c^{(a)}, \mu_s^{(b)})$ for 500 donor pairs and evaluates pixels with the independent real-image classifier. On unrepaired checkpoints, identity follows the style donor on 99.28% of MNIST and 63.92% of CIFAR-10 off-diagonal swaps, while following the semantic donor on 0.06% and 4.72%. At $\delta = 1$, style-following falls to 23.37% and 35.92%, and semantic- following rises to 51.00% and 18.74%, respectively. Full rate matrices, the other-class outcome, and pair-bootstrap intervals are stored in the external-swap artifacts.

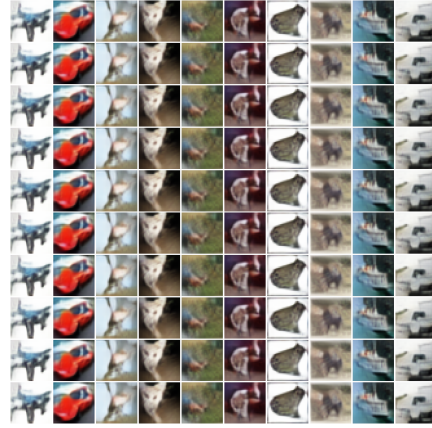
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Latent swap: row = z_c class (content), col = z_s class (style) — mnist_delta0

Latent swap: row = z_c class (content), col = z_s class (style) — cifar10_delta0



(a) MNIST



(b) CIFAR-10

Figure 6: Deterministic mean-swap grids at $\delta = 0$. Rows donate semantic means and columns donate style means. Column-controlled identity is visually strong on MNIST and weaker but still present for several CIFAR-10 classes. Quantitative rates in the text use independent pixel classifiers; the grids are qualitative.

F REPRODUCIBILITY AND CURRENT EVIDENCE BOUNDARY

G EXACT-MARGINAL EXPERIMENT

G.1 CONSTRUCTION AND PROTOCOL

For every replication we draw $n = 2,048$ vectors $z \sim \mathcal{N}(0, I_{128})$ once, then reuse that latent sample while drawing binary labels from logits $[-\kappa z_1, \kappa z_1]$ for $\kappa \in \{0, .5, 1, 2, 4, 8, 16\}$. Thus the population marginal and the finite latent sample are invariant to κ ; only the label mechanism changes. We run 50 replications from base seed 2027, 499 global-MMD permutations, 499 HSIC permutations, and a stratified 60/20/20 linear-probe split. The independent Gaussian reference and the labels use distinct random streams. Plus-one calibration gives minimum $p = 1/500$.

Table 12: Complete synthetic result. Probe values are mean $\pm$ sample standard deviation over 50 replications; rejection columns are empirical rates. The global statistic is repeated across κ within each replication by design.

κ	Probe (%)	$\hat{I}_{LB}$	Global MMD	MMD rej.	HSIC	HSIC rej.	Cond. MMD
0	49.58 $\pm$ 1.86	-.050	-9.47×10^{-7}	.06	.000176	.02	.000006
.5	62.48 $\pm$ 2.58	.031	-9.47×10^{-7}	.06	.000239	1.00	.000130
1	74.49 $\pm$ 2.09	.171	-9.47×10^{-7}	.06	.000314	1.00	.000280
2	83.28 $\pm$ 2.07	.334	-9.47×10^{-7}	.06	.000377	1.00	.000405
4	89.51 $\pm$ 1.45	.437	-9.47×10^{-7}	.06	.000404	1.00	.000455
8	91.82 $\pm$ 1.63	.471	-9.47×10^{-7}	.06	.000413	1.00	.000479
16	92.30 $\pm$ 1.51	.475	-9.47×10^{-7}	.06	.000415	1.00	.000478

The unbiased global MMD can be negative under the null. Three of 50 global tests reject, inside the frozen central 95% prediction interval $[0, 6]$ for Binomial(50, .05). HSIC has Spearman trend 1.0 and conditional MMD trend .964. Seven of eight frozen checks pass. The failed check required mean probe accuracy strictly above 90% at each of $\{4, 8, 16\}$; $\kappa = 4$ reaches 89.51%. No threshold, level, or inequality was changed after observing the result.

The probe effect is essentially stable across an eightfold range in sample size, whereas the rejection rate remains near the nominal 5% level. The follow-up therefore supports a variance-reduction interpretation rather than an effect-size increase with n .

Table 13: Supplementary sample-size sensitivity at $\kappa = 16$. Values are rounded aggregate results from the completed follow-up runs.

n	Probe accuracy (%)	Global-MMD rejection (%)
512	92.0	6.0
1,024	92.2	5.0
2,048	92.3	6.0
4,096	92.5	5.0

H CONTROLLED POSTERIOR-ENTROPY INTERVENTION

The intervention adds a fixed floor in variance space and uses the resulting σ_{eff} in both training and audit. A floor of zero reproduces the historical sampler. Every nonzero setting is retrained for 300 epochs with MNIST model seed 0 and CIFAR-10 model seeds $\{0, 1, 2\}$; the corresponding baseline checkpoints are reused. The CIFAR-10 confirmation levels $\{0, .15, .35\}$ were fixed after both .15 and .35 passed the predeclared MNIST promotion rule. That rule required an entropy increase of at least .75 nats/dimension, probe accuracy at least .20, HSIC rejection, conditional MMD above global MMD, and final reconstruction no more than twice baseline.

Table 14: Complete MNIST entropy sweep. Median standard deviation is the effective posterior value; floor hit refers to the raw log-variance head.

σ_f	H/dim	Med. std	Raw floor	Probe	$\hat{I}_{LB}$	Global	Cond.	Ratio	Gen-ACC	Rec.
0	-3.581	.00674	100.00%	99.76%	2.283	.000579	.016024	27.68	10.15%	.01259
.025	-2.215	.02589	99.16%	99.51%	2.273	.000599	.016352	27.31	10.05%	.01273
.05	-1.553	.05045	99.17%	99.51%	2.273	.000648	.016135	24.88	11.95%	.01315
.15	-.477	.15015	99.77%	98.78%	2.228	.000654	.012044	18.41	37.35%	.01520
.35	.381	.35006	96.50%	95.61%	2.159	.000518	.007516	14.51	93.50%	.01658

Table 15: Seed-0 detail for the fixed CIFAR-10 entropy confirmation.

σ_f	H/dim	Med. std	Probe	$\hat{I}_{LB}$	Global	Cond.	Ratio	Gen-ACC	Rec.
0	-3.581	.00674	61.46%	1.111	.000249	.005789	23.29	12.60%	.05140
.15	-.477	.15015	28.78%	.256	.000743	.003690	4.97	27.35%	.05940
.35	.369	.35006	9.76%	-.115	.000667	.001287	1.93	54.25%	.06476

HSIC attains $p = 1/201$ at every MNIST level and for every CIFAR-10 seed. This outcome is read alongside effect sizes: the three-seed CIFAR-10 .35 probe is only 10.5% and does not support practically recoverable label information even though the kernel test still detects dependence. Raw floor-hit is 100% at all CIFAR levels; the experiment manipulates effective noise rather than claiming that the variance head stops collapsing.

Table 16: Three-seed CIFAR-10 entropy confirmation. Entries are rounded means over model seeds; $p = .00498$ means that every seed reaches the minimum plus-one calibrated HSIC value with 200 permutations.

σ_f	Probe (%)	Global MMD	Cond. MMD	HSIC p	Gen-ACC (%)	Rec. L1
0	68.0	.00030	.0058	.00498	13.0	.052
.15	29.0	.00070	.0037	.00498	27.0	.059
.35	10.5	.00067	.0013	.00498	52.0	.065

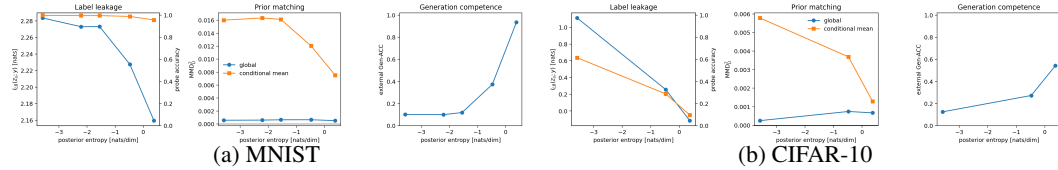


Figure 7: Leakage, marginal/conditional discrepancy, and independent Gen-ACC as effective posterior entropy increases. Curves are single-seed controlled interventions, not confidence intervals over training.

I NATIVE CROSS-MODEL DETAILS

The fixed population contains 40,000 paired training examples and 10,000 paired test examples. Each canonical domain audit uses 1,000 balanced test examples. The independent Rotated-MNIST evaluator is trained only on real images and reaches 96.05% test accuracy. All models use their final 100-epoch checkpoint for seeds 0–2.

Table 17: Per-seed native cross-model results, macro-averaged over the two domains. Probe, content, and retention values are percentages.

Family	Seed	Global	Cond.	Logit	RBF	HSIC domains	Content	Retention
F-CS-WAE	0	.003287	.032431	94.00	97.75	2/2 reject	99.25	12.55
F-CS-WAE	1	.003209	.030989	92.50	96.25	2/2 reject	99.50	11.10
F-CS-WAE	2	.003481	.033365	89.25	96.00	2/2 reject	99.50	13.45
DRIT	0	.259014	.272479	25.00	25.50	2/2 reject	87.50	9.80
DRIT	1	.261287	.274362	29.50	23.25	2/2 reject	90.00	10.00
DRIT	2	.245482	.270487	48.75	48.75	2/2 reject	87.25	9.80
DIVA	0	.0000096	.000268	11.50	12.50	.0746, .0398	96.00	90.25
DIVA	1	-.0000452	.0000288	11.00	11.25	.2687, .0249	97.50	89.50
DIVA	2	-.0000181	.000231	9.50	11.75	.1194, .7264	96.50	88.60

All F-CS-WAE and DRIT domain-level HSIC values equal $1/201$. For DIVA, Holm step-down correction over its six raw tests leaves no rejection. DIVA’s six global-MMD tests also do not reject. Its posterior entropy is approximately 1.36–1.38 nats/dimension; F-CS-WAE is approximately -3.58 and DRIT ranges from -1.88 to -1.57. The cross-model result therefore does not by itself remove posterior entropy as a confound for F-CS-WAE; that role belongs to Appendix H.

J SHAPES3D KNOWN-FACTOR AUDIT

Shapes3D provides 480,000 64×64 images containing every combination of 10 floor hues, 10 wall hues, 10 object hues, 8 scales, 4 shapes, and 15 orientations. The fixed shape-stratified IID split contains 120,000 training, 20,000 validation, and 20,000 test images. F-CS-WAE and the learned C/S VAE use seeds $\{0, 1, 2\}$; the observed-shape oracle uses seed 0. Every model has 64 content and 64 style dimensions and is audited at epoch 100 on 4,096 examples (1,024 per shape). Global

MMD uses 2,048 examples and 500 permutations; factor HSIC uses up to 1,024 examples and 200 permutations. Holm correction is applied separately within each latent view.

Table 18: Completed three-seed Shapes3D replication. Entries are rounded means over seeds $\{0, 1, 2\}$. The observed-shape oracle remains a seed-0 auxiliary control and is reported in the main paper.

Model	$z_s \rightarrow \text{shape}$	$z_c \rightarrow \text{style}$	Cond. shape MMD	Prior shape	Swap shape	Rec. L1
F-CS-WAE	99.0	87.0	.0085	89.0	88.0	.0023
Learned C/S VAE	25.5	9.5	.00004	99.7	99.8	.0029

Table 19: Seed-0 held-out logistic factor probes on stochastic latent samples. Columns follow dataset order. The intended pattern is content high only for shape and style low only for shape.

Model	View	Floor	Wall	Object	Scale	Shape	Orient.
F-CS-WAE	content	99.76	99.88	80.98	68.05	100.00	89.88
	style	100.00	100.00	100.00	95.61	100.00	100.00
Observed-shape cVAE	content	9.76	9.51	9.02	14.27	100.00	8.29
	style	100.00	99.63	99.02	45.49	25.98	39.76
Learned C/S VAE	content	8.90	11.22	8.90	11.95	100.00	5.85
	style	84.02	91.46	88.29	44.15	25.73	42.93

After Holm correction, F-CS-WAE rejects dependence for all six factors in both views. Each control rejects only shape in content and only the five nuisance factors in style. The corresponding shape HSIC raw p -values in style are .597 for the observed-shape cVAE and .826 for the learned model.

Table 20: Seed-0 independent factor-evaluator decoder checks. Posterior swaps keep content from the source and style from a different-shape donor; “target” is source shape for shape and donor value for nuisance factors.

Model	Floor	Wall	Object	Scale	Orient.	Swap shape	Prior shape
F-CS-WAE	100.00	99.90	99.95	80.93	100.00	87.74	89.43
Observed-shape cVAE	99.98	100.00	99.80	95.02	99.93	100.00	99.88
Learned C/S VAE	99.98	99.93	99.85	99.73	99.93	100.00	99.78

All three models pass the frozen competence checks: independent factor evaluator, content-shape probe, reconstruction L1 below .12, reconstruction shape accuracy above .8, and mean reconstructed nuisance-factor accuracy above .7. The observed-shape cVAE supplies content directly from the label; it is an oracle control for the intended split, not a like-for-like representation-learning competitor. The learned content–style VAE is the stronger negative control.

K REPRODUCIBILITY AND EVIDENCE BOUNDARY

The original checkpoint audit uses protocol stage0–1.1.0; all new synthetic, entropy, cross-model, and Shapes3D audits use stage0–1.2.0. Primary artifact roots are:

- runs_diag/synthetic_exact_marginal_paper;
- runs_f/entropy_sweep;
- runs_cross_model/native_v1; and
- runs_shapes3d/factor_audit_v1.

1188 Rounded follow-up values are consolidated in `fcsvae/reported_followup_`
1189 `aggregates_v9.json`. Each final JSON has a SHA-256 sidecar or embeds the
1190 criteria, checkpoint, split, evaluator, source-file, and repository hashes. Synthetic cri-
1191 teria are frozen in `acceptance_criteria.json`, with the composite decision in
1192 `acceptance_result.json`. Entropy summaries record the frozen CIFAR promotion de-
1193 cision. Cross-model reporting is derived from nine native checkpoint audits without using the
1194 legacy proxy artifacts. Shapes3D suite completion records all three models as complete and
1195 competence-passing. The completed sample-size sweep and additional three-seed MNIST, Fashion-
1196 MNIST, CIFAR-10 entropy, and Shapes3D runs are available here as rounded aggregate exports.
1197 We do not reconstruct standard deviations or aggregate p -values from those summaries; seedwise
1198 claims are stated only when the same calibrated decision holds for every run.

1199 The synthetic manifest and several model audits record a dirty worktree. The artifact hashes make
1200 the evaluated outputs immutable, but a camera-ready archive should include the exact source diff or
1201 regenerate from a clean tagged revision. The present scope is 50 synthetic replications at the primary
1202 $n = 2,048$ setting, four sample sizes at $\kappa = 16$, three training seeds per native cross-model family,
1203 three-seed original-image headlines, three CIFAR-10 entropy seeds, three F-CS-WAE/learned-VAE
1204 Shapes3D seeds, and one MNIST entropy and observed-shape-oracle seed. We make no uncertainty
1205 claim beyond those boundaries. Submission source, rendered author blocks, and supplementary
1206 metadata must remain anonymous during review.

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